

**AERO50009 Propulsion and Turbomachinery
Tutorial Sheet 2**

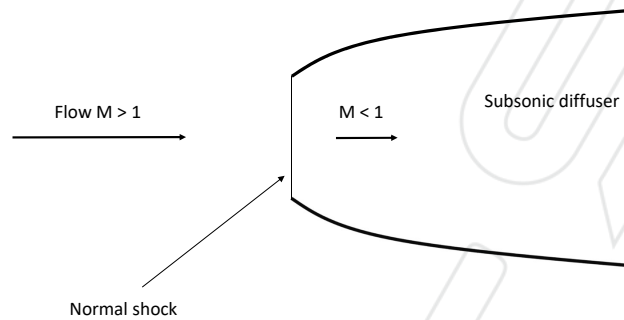


Figure 1: Supersonic diffuser with a normal shock

1. Consider the supersonic diffuser in Fig. 1. In this diffuser the air passes through a normal shock at the inlet cross-section and then it is further slowed down to practically zero velocity in a subsonic part of the diffuser. Assume, that the subsonic part of the diffuser is ideal. For $M=3$, calculate the adiabatic efficiency of this diffuser, η_d .

Use the Compressible Aerodynamics Calculator (also linked from Blackboard) at <http://www.dept.aoe.vt.edu/~devenpor/aoe3114/calc.html> for calculations of the shock.

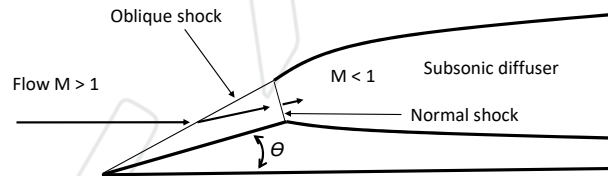


Figure 2: Supersonic diffuser with an oblique and a normal shock

2. Consider the supersonic diffuser in Fig. 2. In this diffuser the air first passes through an oblique shock and then through a normal shock followed by the subsonic part of the diffuser. Assume, that the subsonic part of the diffuser is ideal, and that the normal shock does not cause boundary layer separation. For $M=3$ and the deflection angle $\theta = 5^\circ$, calculate the adiabatic efficiency of this diffuser. Compare with the first one.

3. If truly interested, try to vary θ and see what happens.

4. Compare air intakes of [MiG-19](#) and [Su-27](#) (click the plane names to see the pictures).

$$1. \eta_d = \frac{h_{ods} - h_a}{h_{od} - h_a} = \frac{T_{ods} - T_a}{T_{od} - T_a} = \frac{\frac{T_{ods}}{T_a} - 1}{\frac{T_{od}}{T_a} - 1}$$

$$= \frac{\left(\frac{p_{o2}}{p_1}\right)^{\frac{\delta-1}{\delta}} - 1}{\frac{T_2}{T_1} \frac{T_{o2}}{T_1} - 1} = \frac{\left(\frac{1}{0.0291210}\right)^{\frac{0.4}{1.4}} - 1}{\frac{2.67901234}{0.95679012} - 1} = 0.5760$$

$$2. \eta_d = \frac{\frac{T_{ods}}{T_a} - 1}{\frac{T_{od}}{T_a} - 1}$$

$$\therefore \frac{T_{ods}}{T_a} = \left(\frac{p_{od}}{p_u}\right)^{\frac{\delta-1}{\delta}} = \left(\frac{p_2}{p_1} \frac{p_{o2}}{p_2} \frac{p_{o3}}{p_{o2}}\right)^{\frac{\delta-1}{\delta}}$$

$$= (1.45398306 \frac{1}{0.03974445} 0.40632520)^{\frac{0.4}{1.4}}$$

$$= 2.161453411$$

$$\frac{T_{od}}{T_a} = \frac{T_2}{T_1} \frac{T_3}{T_2} \frac{T_{o3}}{T_3} = 1.11456997 \cdot 2.39624970 \cdot \frac{1}{0.95385225} = 2.799999975$$

$$\therefore \eta_d = 0.645251904$$

3. ...

4. ...